

Sentiment Analysis Report

1. Dataset Description

The dataset used in this project is the *Datafiniti Amazon Consumer Reviews dataset*, which contains customer reviews of Amazon products. The primary feature used for this task is the "reviews.text" column, which includes textual customer feedback.

2. Preprocessing Steps

The following preprocessing steps were applied:

- Removed missing values from the dataset
- Converted text to lowercase
- Removed leading/trailing whitespace
- Tokenised text using spaCy
- Removed stop words (e.g., "the", "is", "and")
- Removed punctuation

These steps helped clean the data and improve the quality of sentiment analysis.

3. Evaluation of Results

The sentiment analysis model was tested on several product reviews. The model successfully classified reviews into:

- Positive
- Negative
- Neutral

Most clearly positive or negative reviews were classified correctly. However, some neutral or ambiguous reviews were harder to classify accurately.

4. Strengths and Limitations

Strengths:

- Easy to implement using spaCy and TextBlob
- Works well for clear sentiment (strongly positive/negative)
- Fast processing of large datasets

Limitations:

- Struggles with sarcasm or complex language
- May misclassify neutral or mixed sentiment reviews
- Depends heavily on the quality of preprocessing
- Less accurate than advanced deep learning models